

POMDP Formulation Approximations

POMDP Computational Complexity

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Sad facts 🥹

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- Infinite horizon POMDPs are *undecidable*

POMDP Computational Complexity

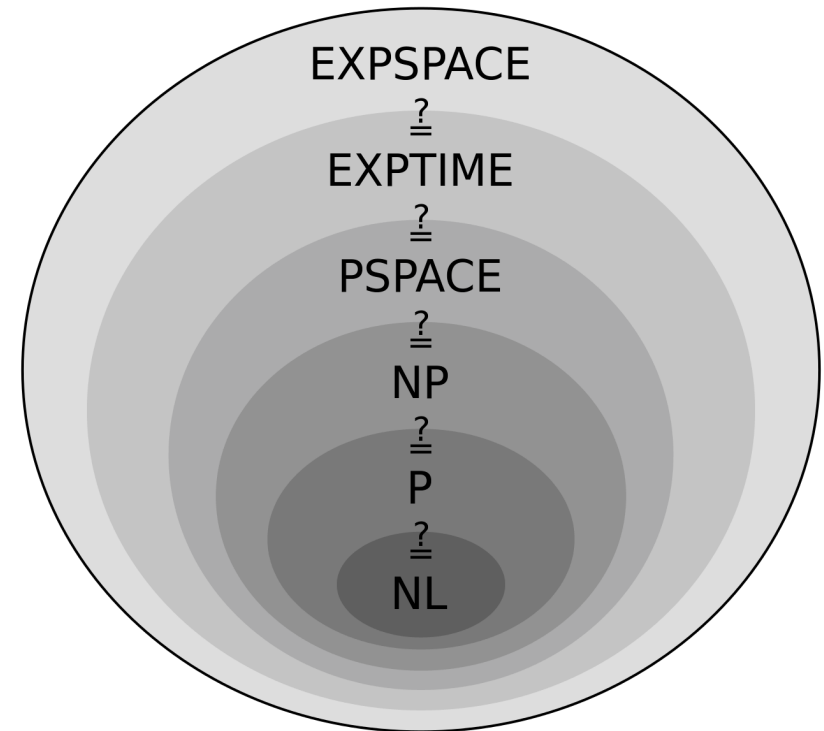
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- Infinite horizon POMDPs are *undecidable*
- Finite horizon POMDPs are *PSPACE Complete*
 - Among the hardest problems that can be solved using a polynomial amount of space

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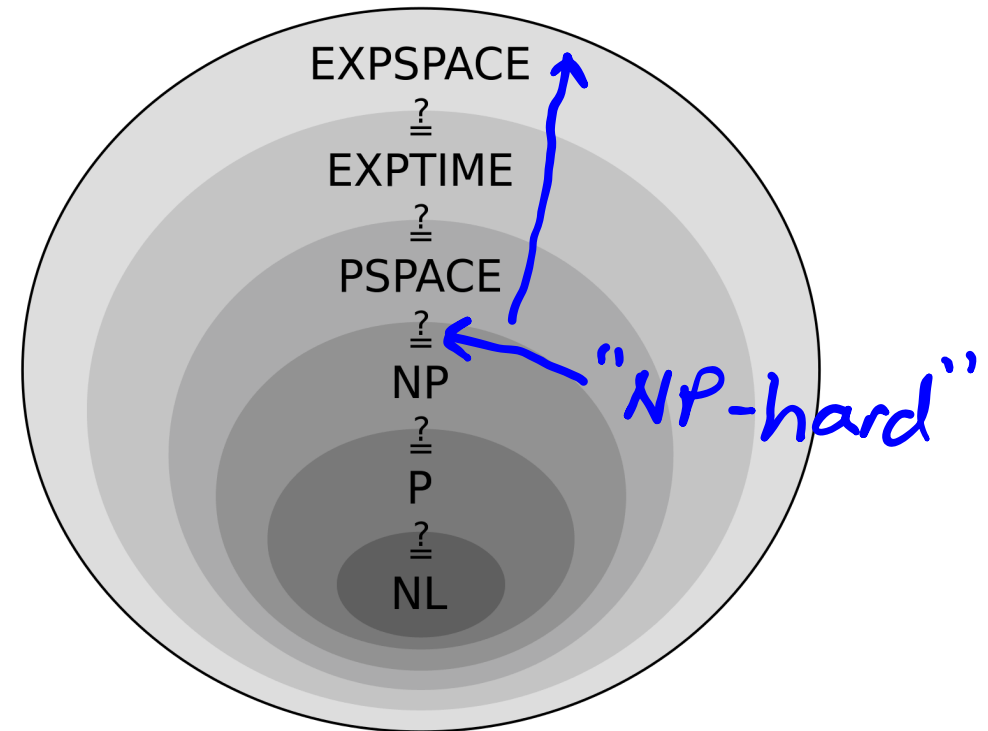
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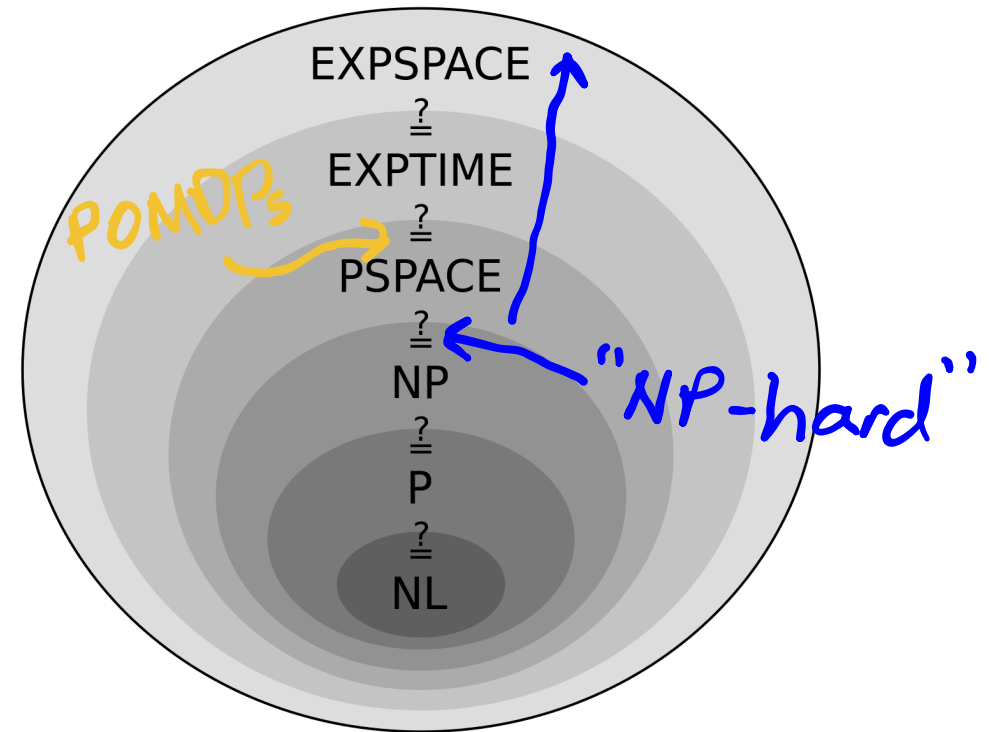
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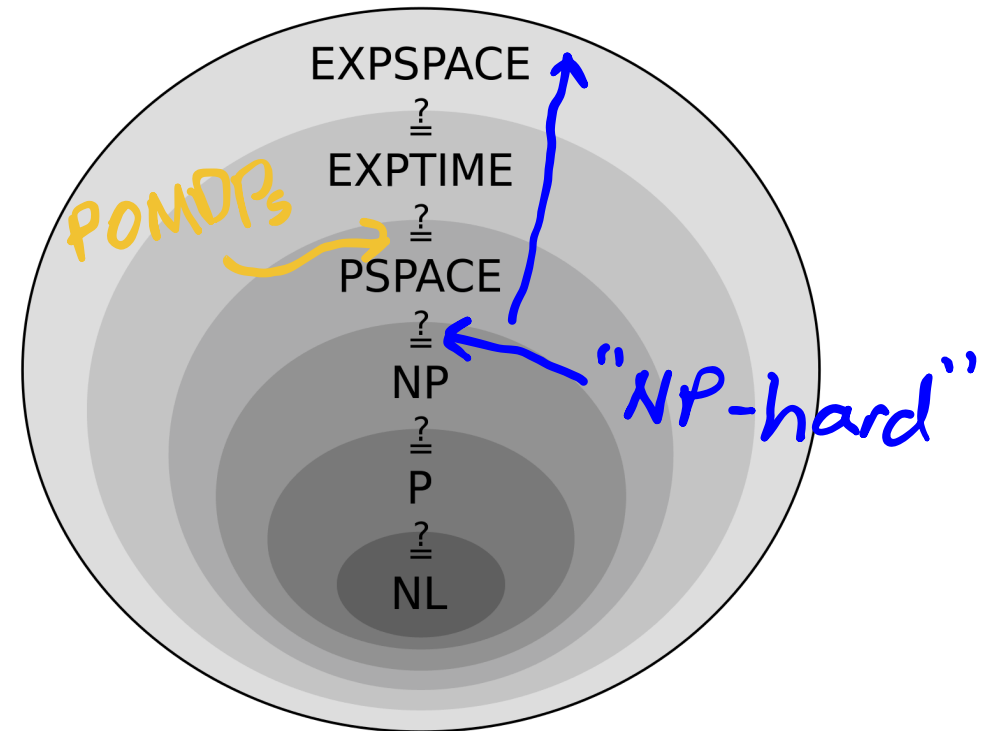
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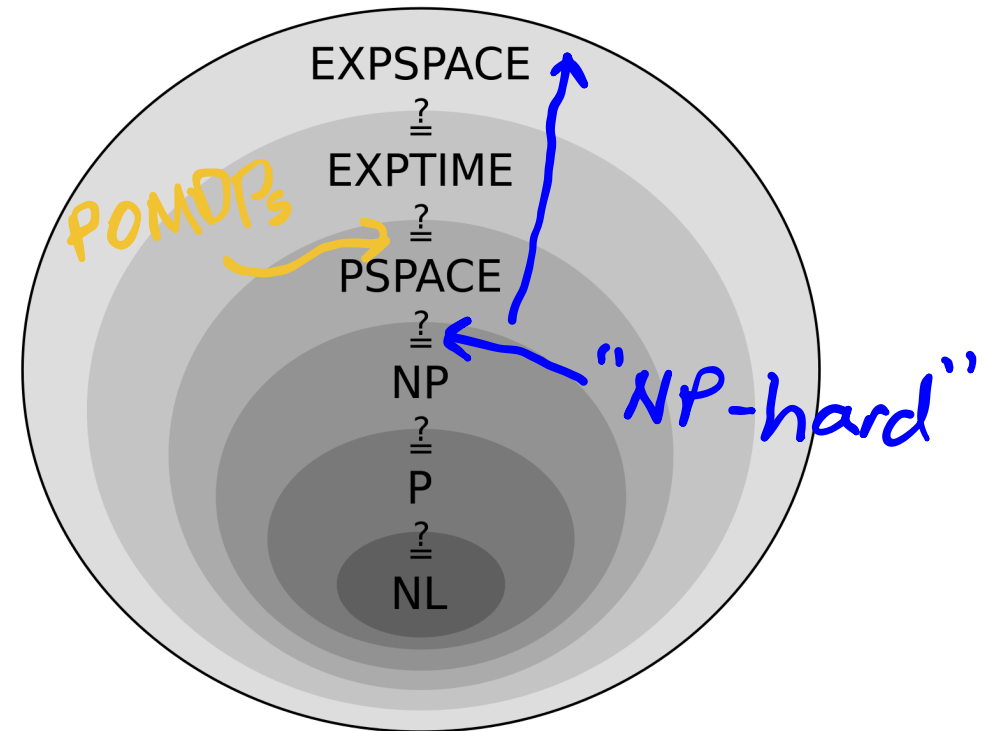
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 - Any algorithm that can solve a general POMDP will have exponential complexity (we think)



Approximate POMDP Solutions

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Numerical Approximations

(approximately solve original problem)

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(approximately solve original problem)



Offline

Approximate POMDP Solutions

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(approximately solve original problem)



Offline

SARSOP

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(approximately solve original problem)



Offline

SARSOP



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Offline

SARSOP



Online

(Next time)

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Formulation Approximations

(solve a slightly different problem)

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Today!

POMDP Objective

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$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

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Certainty Equivalent

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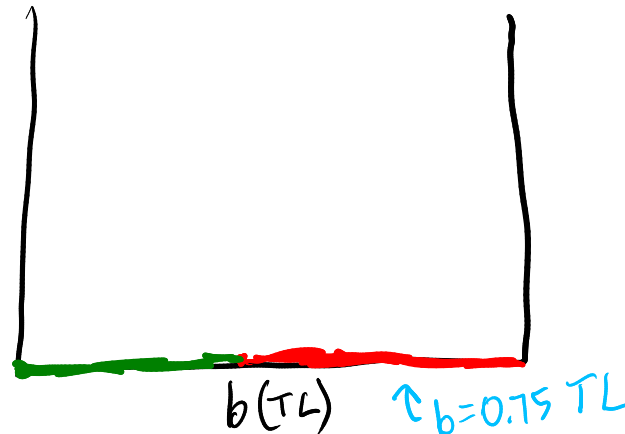
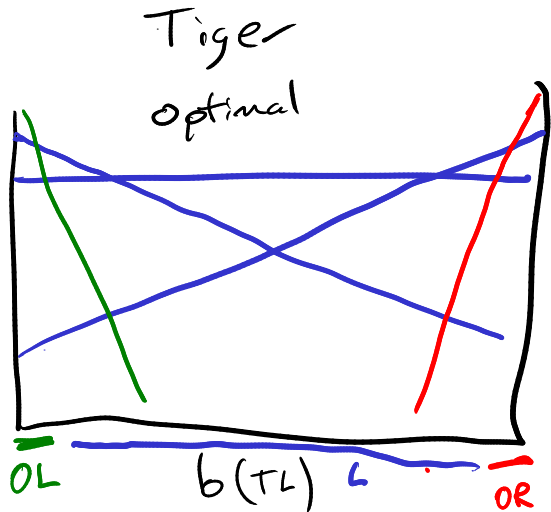
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$\hat{s}$ choices: mean: $\mathbb{E}_{s \sim b}[s]$
mode: $\operatorname{argmax}_{s \in S} b(s)$

$$\pi_{\text{CE}}(b) = \pi_s(\hat{s}(b))$$

π_s : optimal policy for s
without any noise

$$b' = \tau(b, a, o)$$

In a looser sense CE can refer to other choices of π_s such as optimal MDP solution

Certainty Equivalent

Optimal for Linear-Quadratic-Gaussian (LQG)
(Analogous to LQR MDP)

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LQG POMDP

$$\mathcal{S} = \mathbb{R}^n, \mathcal{A} = \mathbb{R}^m, \mathcal{O} = \mathbb{R}^p$$

$$T(\mathbf{s}' \mid \mathbf{s}, \mathbf{a}) = \mathcal{N}(\mathbf{s}' \mid \mathbf{T}_s \mathbf{s} + \mathbf{T}_a \mathbf{a}, \Sigma_s)$$

$$O(\mathbf{o} \mid \mathbf{s}') = \mathcal{N}(\mathbf{o} \mid \mathbf{O}_s \mathbf{s}', \Sigma_o)$$

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$$\boldsymbol{\mu}_p \leftarrow \mathbf{T}_s \boldsymbol{\mu}_b + \mathbf{T}_a \mathbf{a}$$

$$\Sigma_p \leftarrow \mathbf{T}_s \Sigma_b \mathbf{T}_s^\top + \Sigma_s$$

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$$\mathbf{K} \leftarrow \Sigma_p \mathbf{O}_s^\top (\mathbf{O}_s \Sigma_p \mathbf{O}_s^\top + \Sigma_o)^{-1}$$

$$\mu_b \leftarrow \mu_p + \mathbf{K}(\mathbf{o} - \mathbf{O}_s \mu_p)$$

$$\Sigma_b \leftarrow (\mathbf{I} - \mathbf{K} \mathbf{O}_s) \Sigma_p$$

Kalman
Filter

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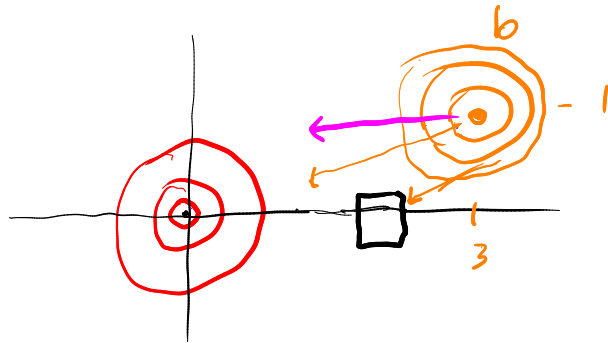
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$$O(o | s') = \mathcal{N}(o | O_s s', \Sigma_o)$$

$$R(s, a) = -\underline{s^\top R_s s} - a^\top R_a a$$



π_{LQG}

$$b(s) = \mathcal{N}(s | \mu_b, \Sigma_b)$$

$$\mu_p \leftarrow T_s \mu_b + T_a a$$

$$\Sigma_p \leftarrow T_s \Sigma_b T_s^\top + \Sigma_s$$

$$K \leftarrow \Sigma_p O_s^\top (O_s \Sigma_p O_s^\top + \Sigma_o)^{-1}$$

$$\mu_b \leftarrow \mu_p + K(o - O_s \mu_p)$$

$$\Sigma_b \leftarrow (I - K O_s) \Sigma_p$$

Kalman Filter

$$\pi_{LQG}^*(b) = -K_{LQR} \mu_b$$

Linear Policy

QMDP

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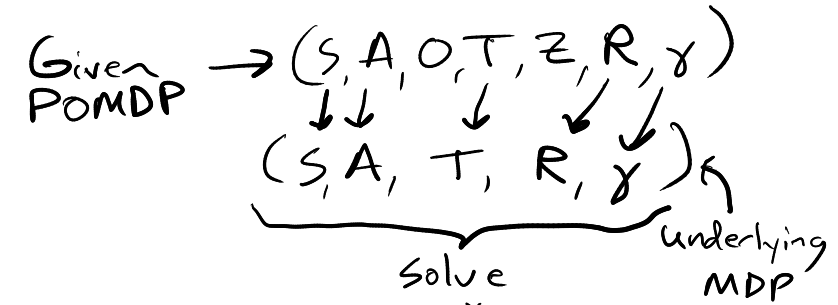
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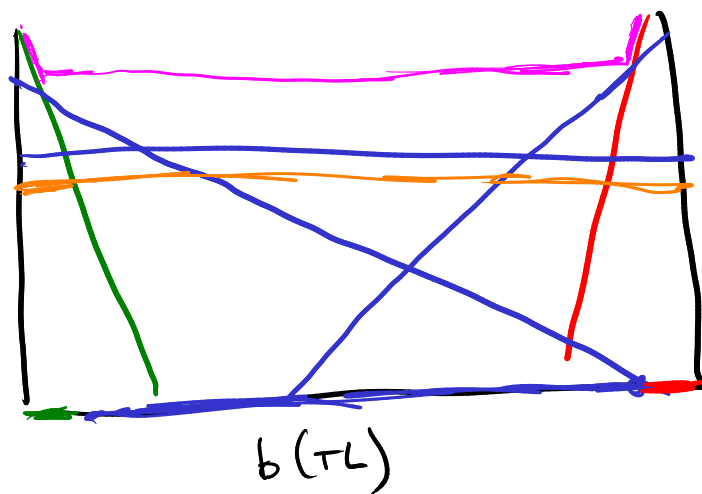


$$\pi_{\text{QMDP}}(b) = \operatorname{argmax}_{a \in A} \mathbb{E}_{s \sim b} [Q_{\text{MDP}}^*(s, a)]$$

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Example: Tiger POMDP with Waiting

Terminates when door opened



QMDP pseudo- α -vectors are an upper bound on optimal true α -vectors

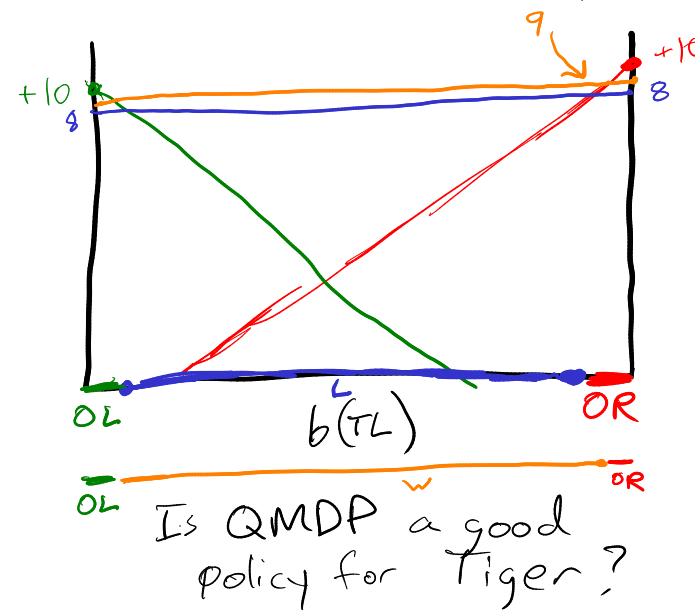
Wait

50-50 observation
0 immediate reward

$$\arg\max_a E_{s \sim b} [Q_{MDP}(s, a)]$$

$$E_{s \sim b} [Q_{MDP}(s, a)] = b(TL) Q_{MDP}(TL, a) + (1 - b(TL)) Q_{MDP}(TR, a)$$

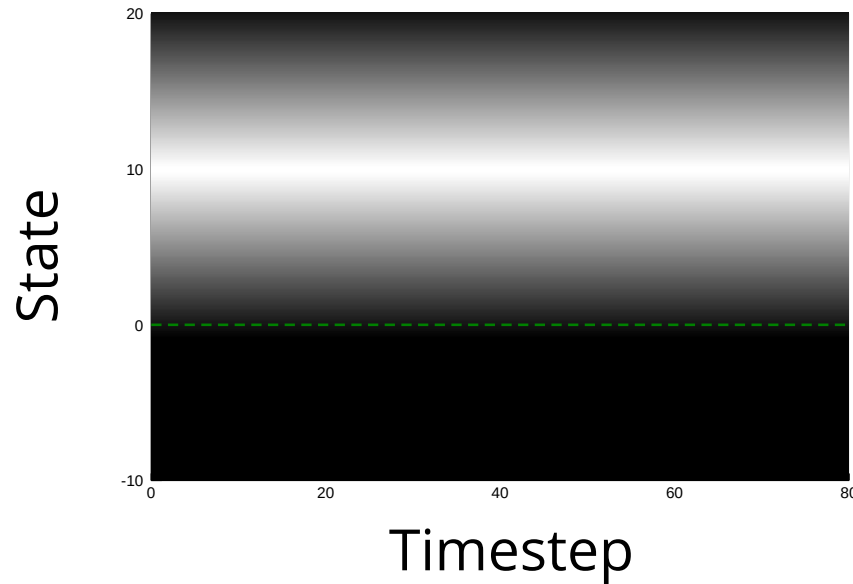
defines some pseudo- α -vector



s	a	Q_{MDP}
TL	OL	-100
TL	OR	+10
TR	OL	+10
TR	OR	-100
*	L	$-1 + \gamma 10 = 9$
*	W	$0 + \gamma 10 = 9$

QMDP is bad at costly information gathering + long-lasting uncertainty

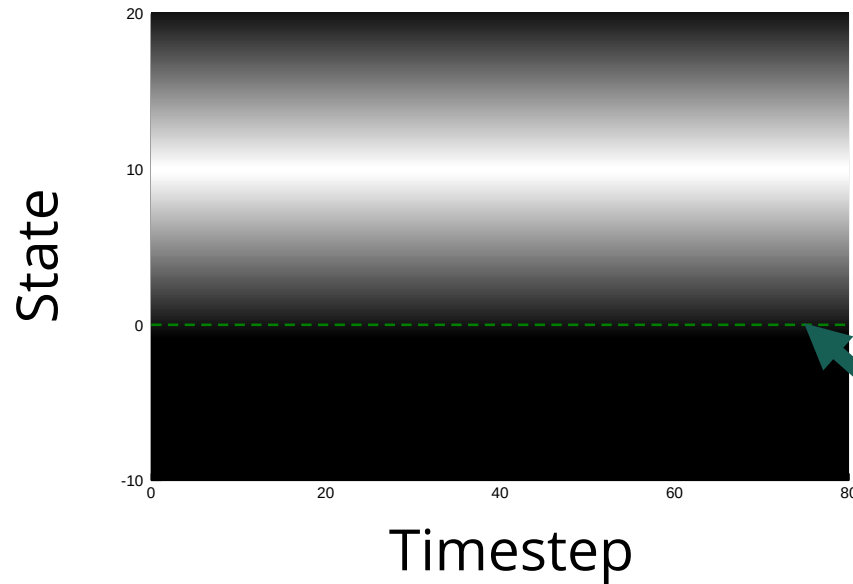
POMDP Example: Light-Dark



$$\begin{aligned}\mathcal{S} &= \mathbb{Z} & \mathcal{O} &= \mathbb{R} \\ s' &= s + a & o &\sim \mathcal{N}(s, s - 10) \\ \mathcal{A} &= \{-10, -1, 0, 1, 10\}\end{aligned}$$

$$R(s, a) = \begin{cases} 100 & \text{if } a = 0, s = 0 \\ -100 & \text{if } a = 0, s \neq 0 \\ -1 & \text{otherwise} \end{cases}$$

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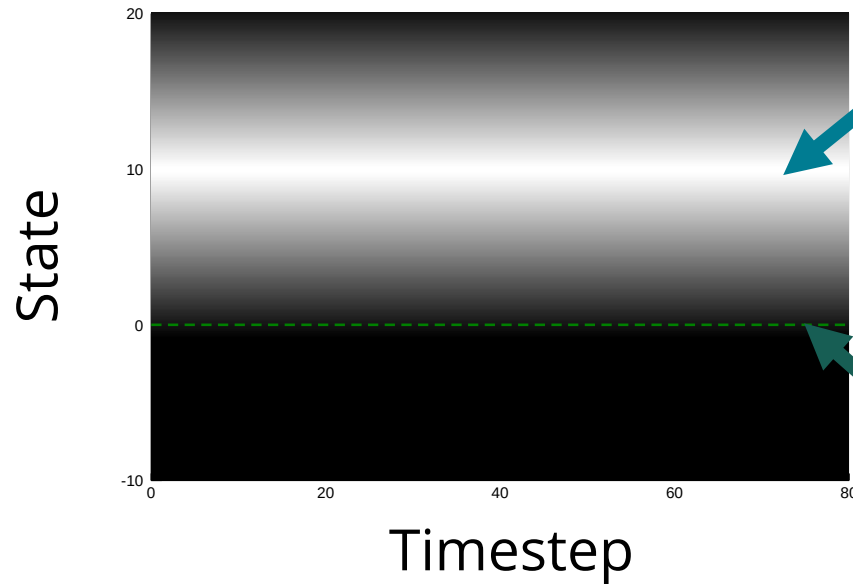
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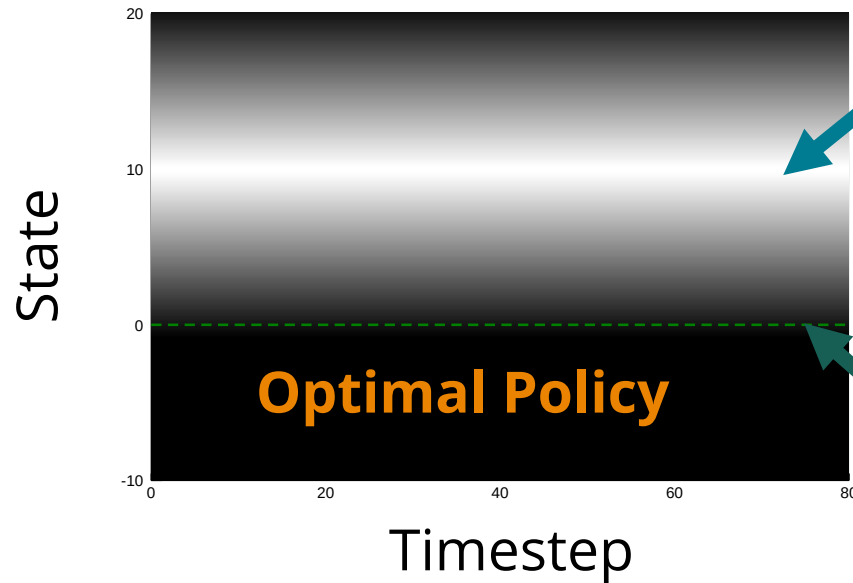
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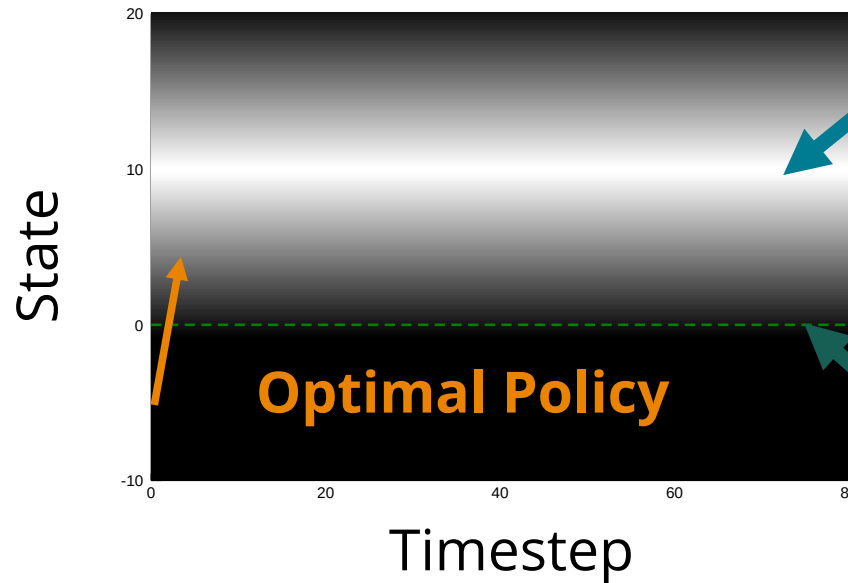
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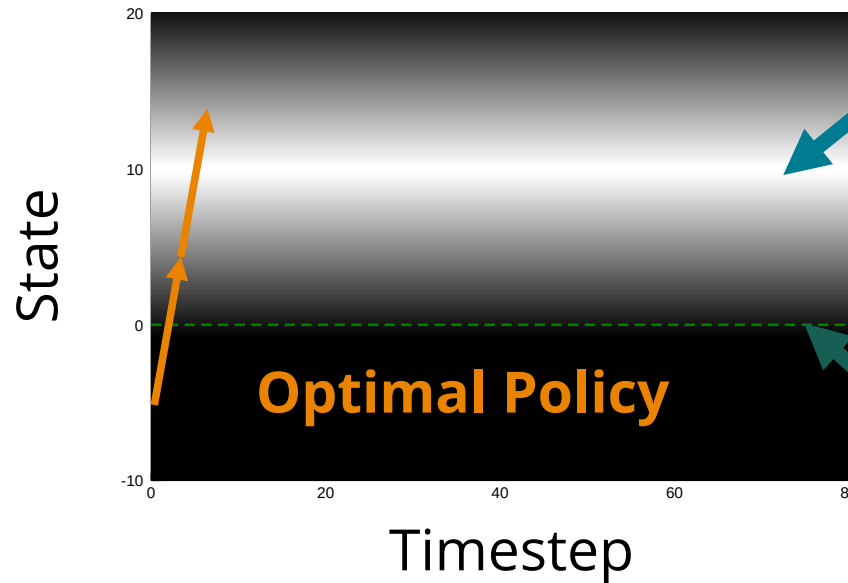
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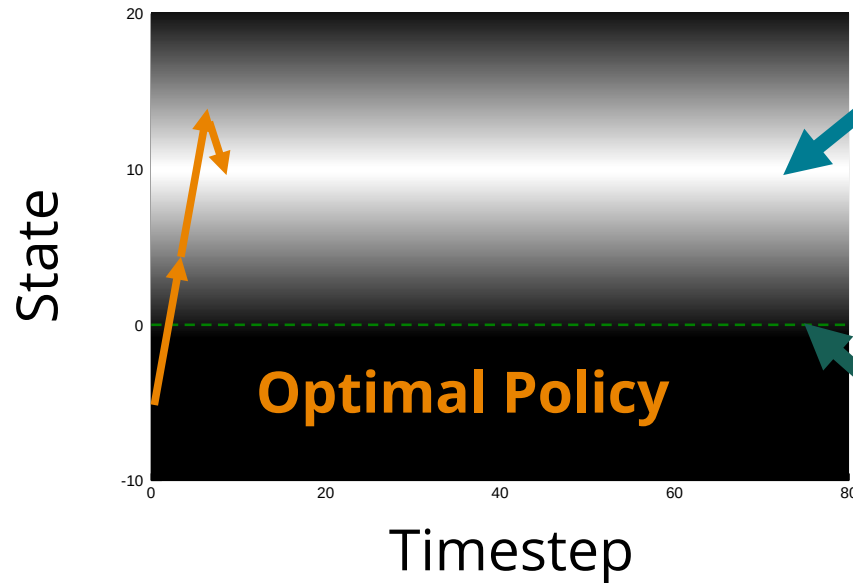
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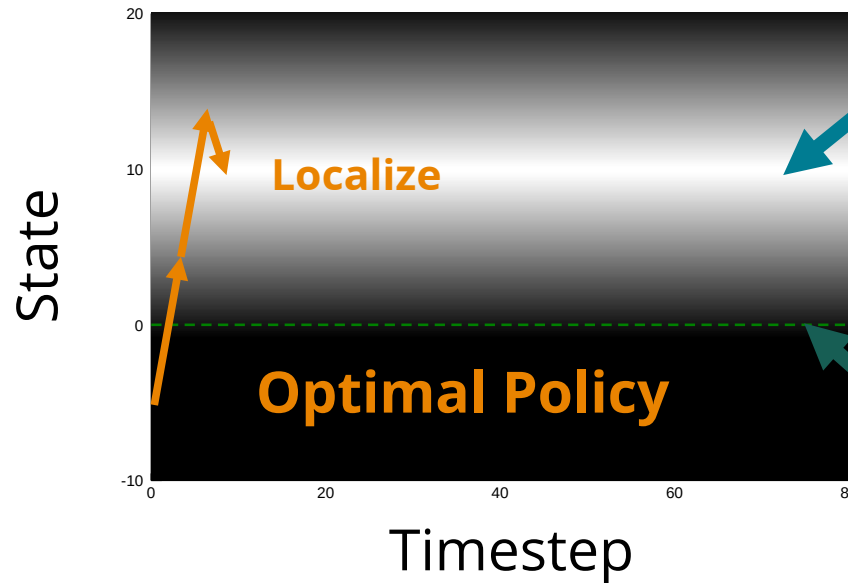
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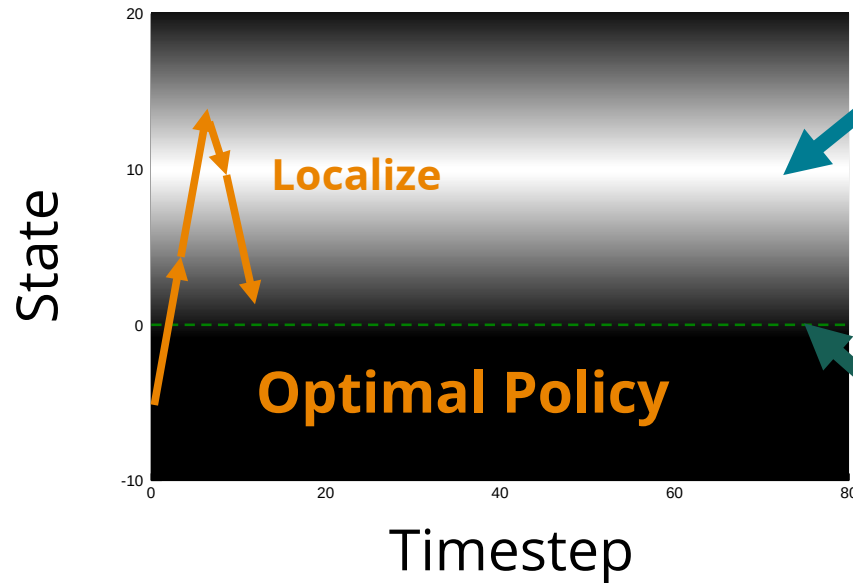
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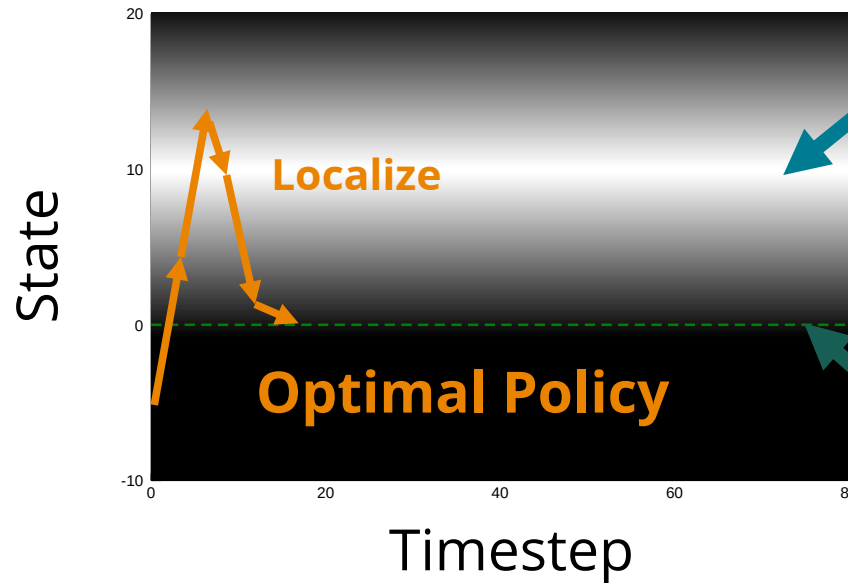
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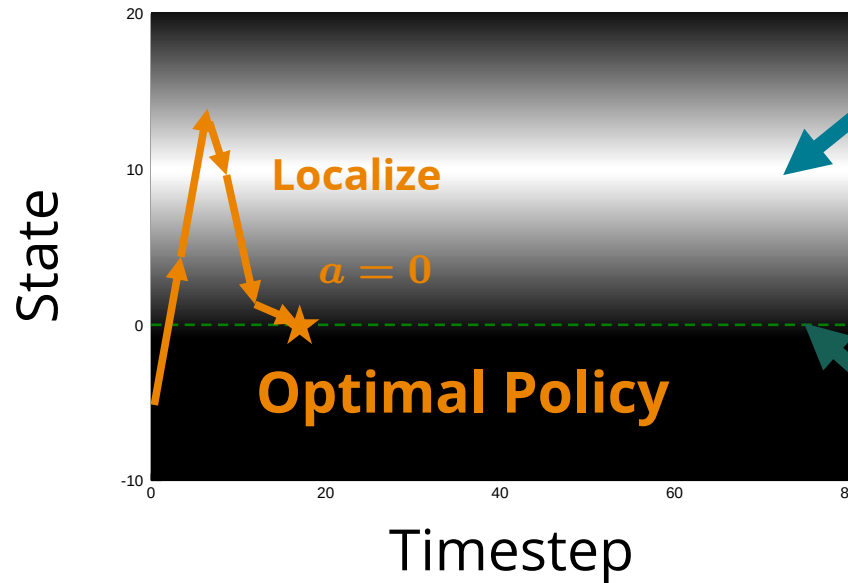
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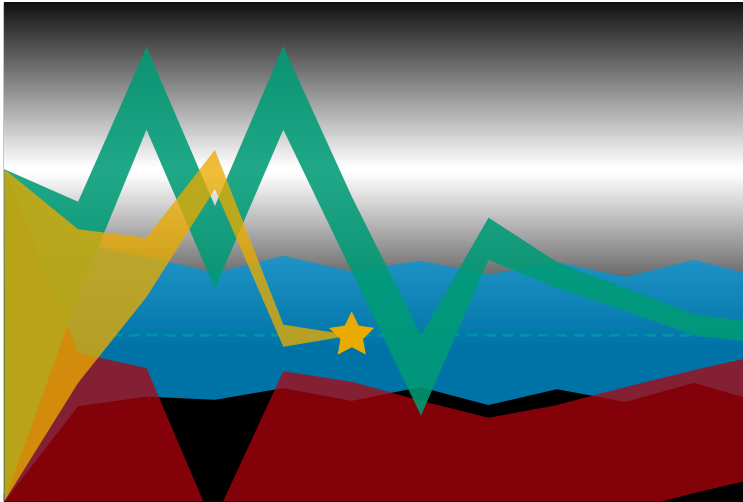
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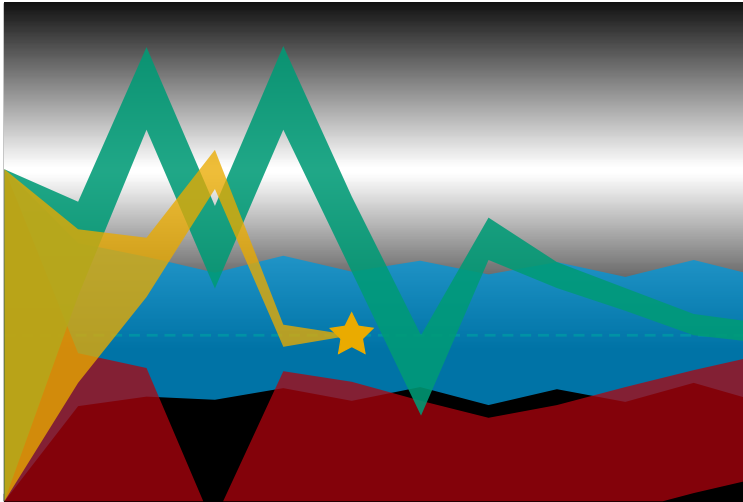
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POMDP

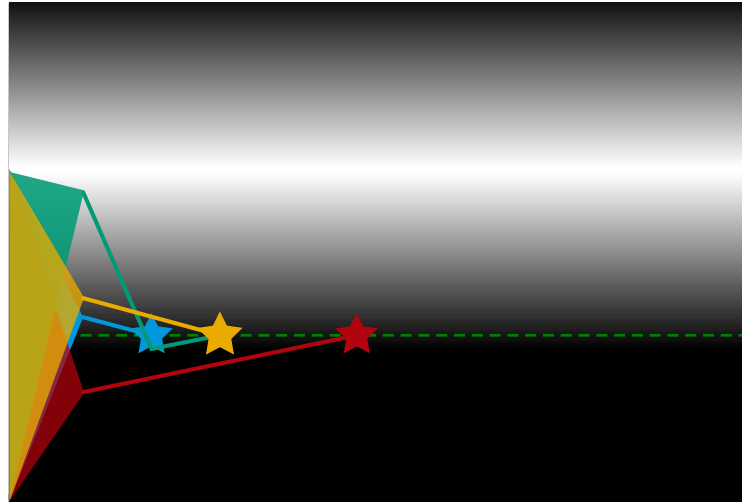
"Imagination"



POMDP "Imagination"

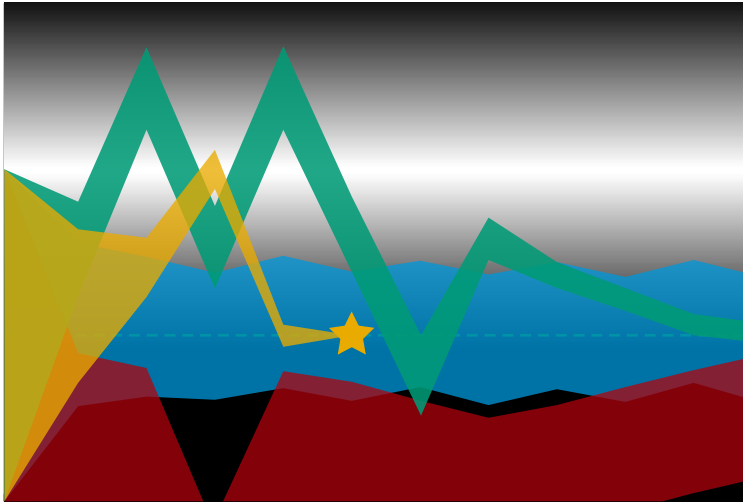


QMDP "Imagination"

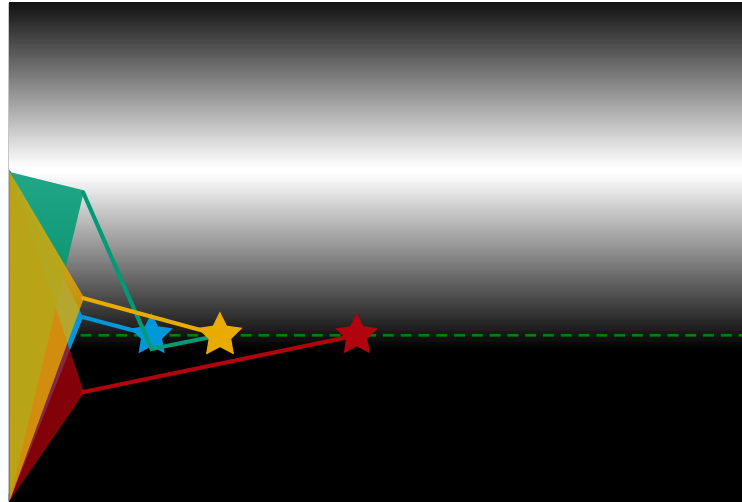


Same as **full observability**
on the next step

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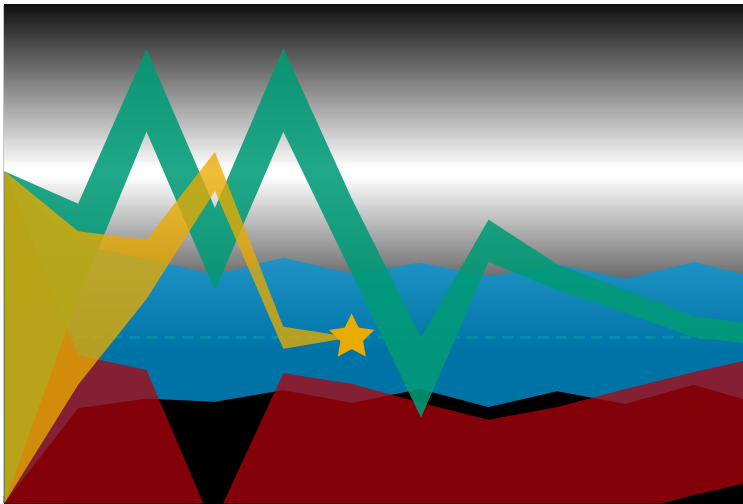
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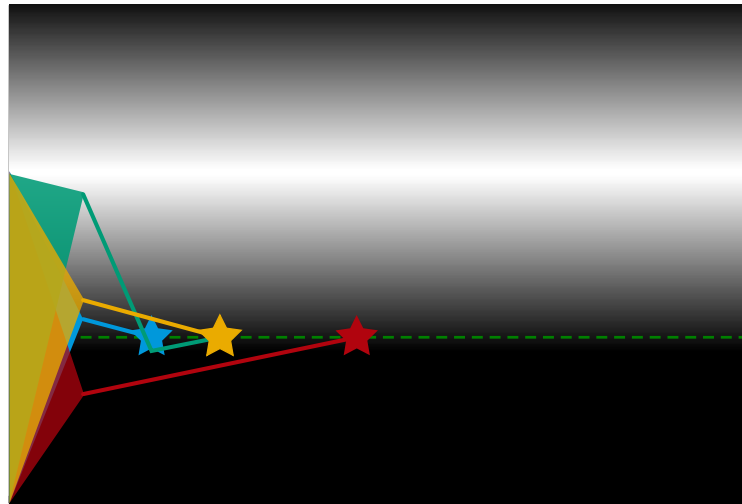
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QMDP Behavior (Break)

POMDP "Imagination"

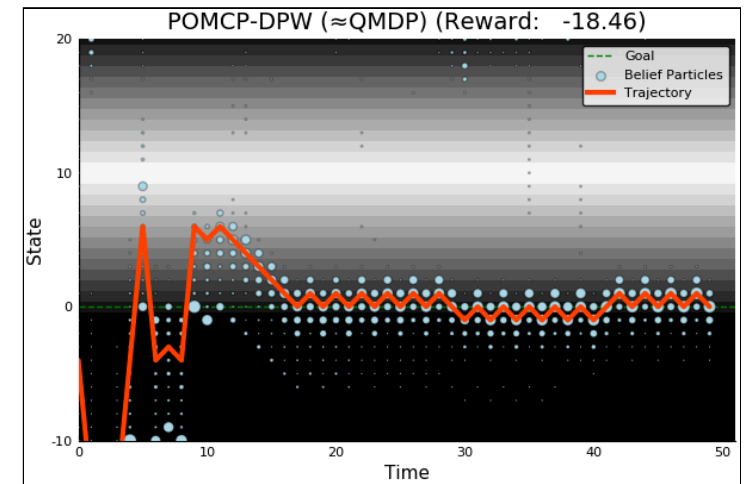


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QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

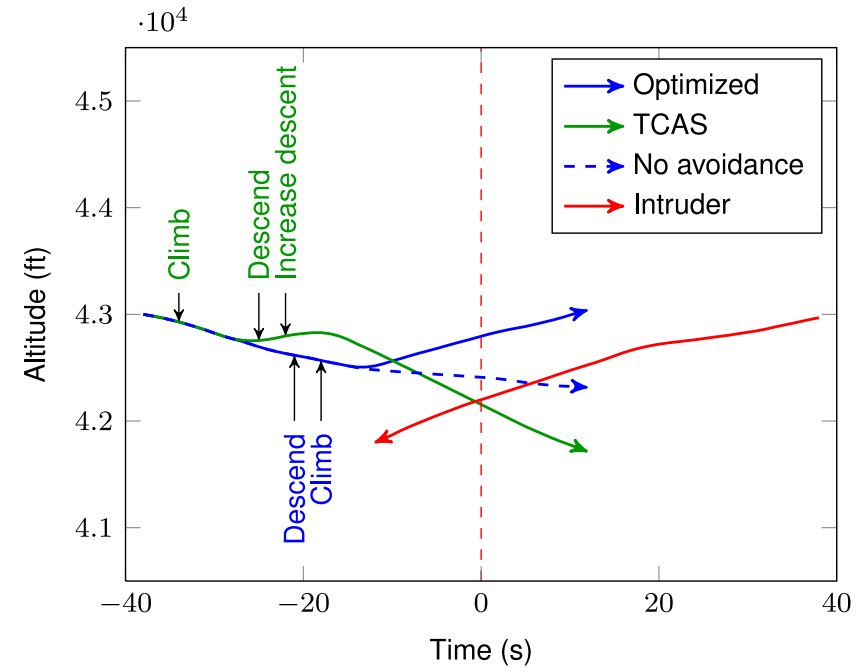
Two cases where it may not work well:

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Used for ACAS X

[Kochenderfer, 2011]



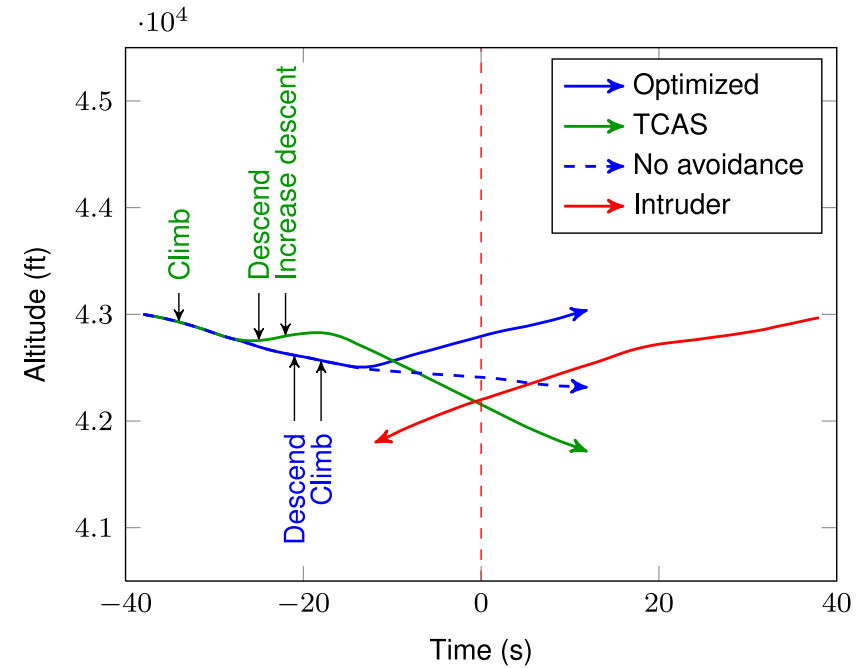
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[Kochenderfer, 2011]



Two cases where it may not work well:

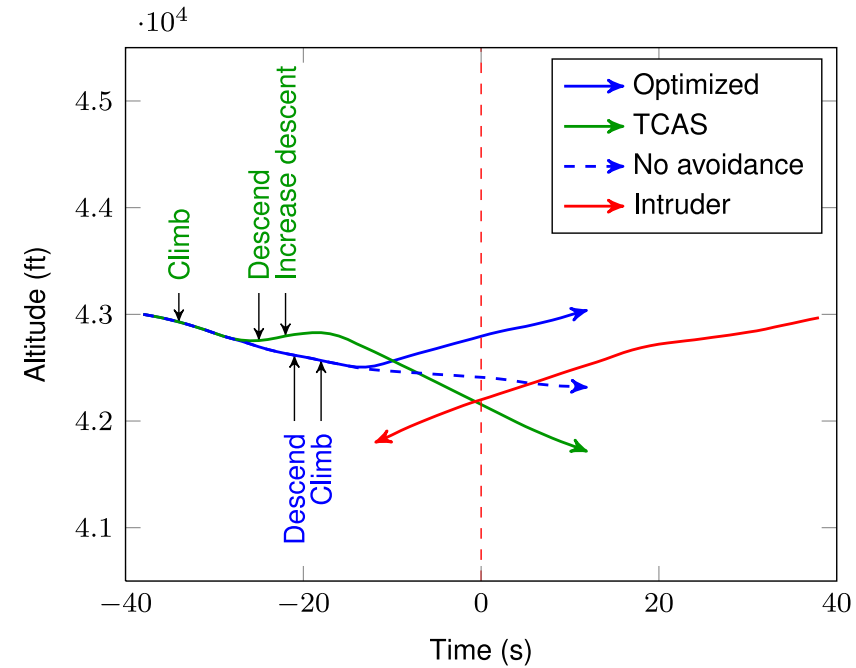
- *Costly* information gathering

QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

Used for ACAS X

[Kochenderfer, 2011]



Two cases where it may not work well:

- *Costly* information gathering
- Long-lived uncertainty

Hindsight Optimization

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

K scenarios

pre sample $\underline{w}_t^k$

$$Q_{HS}(s, a) = R(s, a) + \gamma \frac{1}{K} \sum_{k=1}^K \left(\max_{a_{1:T}^k} \sum_{t=1}^T \gamma^{t-1} R(s_t^k, a_t^k) \right)$$

subject to $s_{t+1}^k = G(s_t^k, a_t^k, w_t^k)$

$$\pi_{HS}(b) = \operatorname{argmax}_a \mathbb{E}_{s \sim b} [Q_{HS}(s, a)]$$

$$b' = \tau(b, a, o)$$

Some authors $QMDP \Leftrightarrow H.O.$

k-Markov

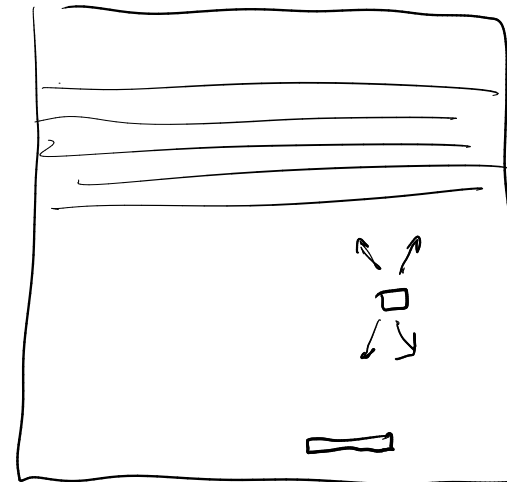
POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$b' = \tau(b, a, o)$$

solve an MDP where

$$s_t = \underbrace{[o_t, o_{t-1}, \dots, o_{t-k+1}]}_k$$



Open Loop

A.K.A. Open Loop Feedback Control (OLFC)

POMDP Objective

$$\pi^* = \operatorname{argmax}_{\pi: B \rightarrow A} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right]$$

$$\pi_{\text{OLFC}}(b) = \operatorname{argmax}_{a_0} \underbrace{\left(\max_{a_{1:T}} \mathbb{E}_{s_0 \sim b} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right] \right)}_{\text{Open-loop}}$$

Feedback

$$b' = \tau(b, a, o)$$

